FITS Extractor

User Manual

Extractor v1.4.3 | Stacker v1.0.1

Updated manual for batch spectrum extraction, continuum handling, normalization, S/N calculation, and stacking of extracted spectra.

1. Overview

FITS Extractor is a small Python GUI application for extracting one-dimensional spectra from FITS binary tables. The launcher provides two applications: FITS Extractor v1.4.3 for reading FITS binary tables and writing 1D spectrum files, and FITS Extractor Stacker v1.0.1 for stacking spectra that have already been extracted into primary-HDU spectrum files.

This manual assumes that the user is already familiar with FITS files, binary table extensions, and the meaning of wavelength, flux, flux error, continuum, and status columns.

2. Installation and requirements

Install the Python dependencies in the project directory with:

```
pip install -r requirements.txt
```

The current dependency list should include the runtime packages used by both the extractor and the stacker:

- **astropy** - FITS I/O and physical units
- **numpy** - array operations
- **scipy** - interpolation, filtering, and Savitzky-Golay smoothing
- **matplotlib** - static plotting and optional previews
- **tkinterdnd2** - drag-and-drop support for the Tkinter GUI
- **specutils** - spectrum containers and flux-conserving resampling for stacking
- **pandas** - tabular wavelength/flux handling in the stacker
- **plotly** - interactive HTML output plots in the stacker

> ***Note:*** *Tkinter itself is part of many Python installations, but on some systems it must be installed through the operating-system package manager rather than pip.*

> ***Note:*** *If you want to build an executable using fits_extractor_init.spec, PyInstaller is also required. The spec file also collects data from asdf, astropy, specutils, pandas, and tkinterdnd2 for packaging.*

3. Start the application

Run the launcher from the project directory:

```
python fits_extractor_init.py
```

The launcher opens a graphical selection window. Choose either "Run FITS Extractor 1.4.3" or "Run FITS Extractor Stacker 1.0.1".

4. FITS Extractor v1.4.3

The extractor reads spectra from FITS binary tables, optionally filters and resamples the data, optionally subtracts a continuum, optionally normalizes the flux, optionally calculates S/N values, and writes new 1D FITS primary-HDU files.

4.1 Load FITS files or directories

Drag and drop either a single FITS file or a directory into the drop area. The GUI displays the path, the total number of loaded files, and the combined file size. When a directory is dropped, all files found recursively in that directory are added to the internal file list. Non-FITS files are not removed immediately; they are skipped later during compatibility checks or extraction.

> ***Note:*** *Only files ending in .fits or .FITS are treated as compatible input files.*

4.2 Load and select binary-table column keys

Click "Load Key Values" after loading files. The extractor scans the loaded FITS files and collects available binary table column names from extension 1. The detected column names populate the dropdown boxes on the right-hand side.

Select the columns corresponding to:

- **wavelength** - default expected key: WAVE
- **flux** - default expected key: FLUX
- **flux error** - default expected key: ERR_FLUX

Optional columns can be enabled separately:

- **Use Continuum Key** - enables selection of a continuum column, default expected key: CONTINUUM
- **Use Status Key** - enables selection of a status column, default expected key: STATUS

> ***Important:*** *The same selected key names are applied to all loaded files. Files that do not contain the selected columns are skipped during extraction.*

4.3 STATUS filtering

When "Use Status Key" is enabled, the selected status column is expected to contain values corresponding to each flux and error value. Values equal to 1 are kept unchanged. Values not equal to 1 are replaced with 0 in both the flux array and the error array.

> ***Important:*** *Status filtering does not remove data points from the wavelength grid. It sets rejected flux and error values to zero, which can affect interpolation and S/N calculation.*

4.4 Interpolation

The "Interpolate Values" option is enabled by default. When active, the extractor creates a uniformly spaced wavelength grid using numpy.linspace from the minimum to the maximum wavelength value with the same number of samples as the original data. Flux and flux-error values are interpolated linearly onto that grid. If a continuum key is enabled, continuum values are interpolated as well.

If interpolation is disabled, the raw wavelength, flux, and error arrays are used. The output FITS header still requires a constant CDELT1 value, so the stored wavelength step is calculated from the first and last wavelength values and the array length. For irregular wavelength sampling, interpolation is recommended.

4.5 Continuum subtraction

The "Subtract Continuum" option is enabled by default. In v1.4.3 there are two active continuum-subtraction paths:

- **Provided continuum column:** If "Use Continuum Key" is enabled and a continuum column is selected, the extractor subtracts the provided continuum values from the flux values. If interpolation is active, the continuum is interpolated before subtraction.
- **Estimated continuum:** If no continuum key is enabled, the extractor estimates the continuum internally using an iterative masked Savitzky-Golay approach. The current extraction call uses window_length=601, polyorder=3, med_width=401, peak_sigma=5.0, mask_width_pix=60, and niter=3.

> ***Note:*** *Continuum-subtraction parameters are written to the output FITS headers only when the continuum is estimated internally. If provided continuum values are used, no Savitzky-Golay parameter keywords are written.*

> ***Important:*** *The CONTINUUM key is only used for subtraction when both "Use Continuum Key" and "Subtract Continuum" are enabled. It is not used for continuum normalization.*

4.6 Normalization

When "Perform Normalization" is enabled, the extractor performs maximum-flux normalization after continuum handling. It finds the index of the maximum flux value and calculates a representative maximum as the median within a small window of +/-5 pixels around that maximum. Flux values are divided by this representative maximum and flux errors are propagated.

The normalization status is encoded in the output file name as normX:

- **norm0** - normalization completed successfully
- **norm1** - normalization was cancelled because of an unknown error
- **norm2** - normalization was cancelled because the representative maximum flux was negative

> ***Important:*** *Continuum normalization and cnormX output names are not active in v1.4.3. The active normalization mode is maximum-flux normalization only.*

4.7 S/N calculation

When "Calculate S/N Values" is enabled, the extractor calculates S/N values as normalized_or_final_flux divided by normalized_or_final_error. If this option is active, an additional S/N FITS file is written for each successfully extracted input file.

4.8 Plot extracted spectra

When "Plot Extracted Spectra" is enabled, a matplotlib preview is shown for each successfully extracted spectrum. The preview contains the final flux curve and flux-error bars. The extraction workflow pauses until the plot window is closed.

4.9 Run extraction

Click "Extract .fits Spectra". Each file is processed sequentially. For each file, the extractor:

1. checks whether the file extension is .fits or .FITS;
2. opens the FITS file and reads the primary header and binary table extension 1;
3. reads wavelength, flux, and flux-error columns using the selected key values;
4. optionally reads continuum and status columns;
5. optionally applies STATUS filtering;
6. flattens single-row vector columns where applicable;
7. optionally interpolates wavelength, flux, error, and continuum arrays;
8. optionally subtracts either a provided or internally estimated continuum;
9. optionally performs maximum-flux normalization;
10. optionally calculates S/N values;
11. writes spectrum, error, and optionally S/N FITS files.

4.10 Extractor output files

The extractor creates one output folder per application run. The folder name is generated when the module is loaded and follows this pattern:

```
extracted_YYYYMMDD-HHMMSS/
```

For each successfully extracted file, the extractor always writes:

- ***_spec.fits** - the extracted final flux spectrum
- ***_err.fits** - the propagated or final flux-error spectrum

If S/N calculation is enabled, it additionally writes:

- ***_sn.fits** - S/N values calculated as final flux divided by final error

The output file name is constructed from:

```
<input-name>_<OBJECT>_<five-character-id>_<option-tags>_<type>.fits
```

Possible option tags are:

- **interp** - interpolation was enabled
- **stat** - STATUS filtering was enabled
- **cntrm** - continuum subtraction was enabled
- **normX** - normalization was enabled, with X indicating the normalization status

The spectrum and error FITS headers include, where available: OBJECT, CUNIT1, CRPIX1, CRVAL1, CDELT1, CRDER1, CTYPE1, and CONTSUB. If the continuum was estimated internally, additional PWINLEN, PPOL_ORD, PMEDWID, PPEAKSIG, PMASKWID, and PNITER keywords are written.

4.11 Skipped files and common console messages

If an input file cannot be processed, no output files are created for that file. Common messages are:

- **warn: File is not compatible!** The input file extension is not .fits or .FITS.

- **err: File is not loading correctly!** The FITS file could not be opened, for example because it is corrupted or not a valid FITS file.
- **err: File throws error when extracting!** The FITS structure or selected column keys do not match the expected binary-table layout.
- **err: No BinTable extension found!** The FITS file does not contain a valid binary-table structure.

5. FITS Extractor Stacker v1.0.1

The stacker combines already extracted 1D spectrum FITS files. It expects spectra stored in the primary HDU, with wavelength information described by FITS header keywords such as CRVAL1 and CDELT1.

5.1 Load spectra for stacking

Drag and drop a directory or multiple FITS files into the stacker drop area. The stacker displays the number of loaded files and the total size. At least two files are required; attempting to stack only one spectrum produces an error message.

> ***Important:*** *The stacker is intended for spectra that have already been extracted into primary-HDU 1D spectrum files. It does not read wavelength, flux, or error columns from binary tables.*

5.2 Stacker options

The GUI currently exposes one option: "Save individual spectra". If enabled, the stacker saves a PNG and an interactive HTML plot for each input spectrum before stacking.

The current GUI call uses the following fixed internal settings:

- **specUnit="AA"** - wavelength unit is Angstrom
- **fluxUnit="erg cm-2 s-1 AA-1"** - flux unit used for the Spectrum object
- **binfactor=1** - the resampling step is not multiplied beyond the calculated base step
- **z=0** - no redshift correction is applied through the GUI
- **statistic="mean"** - spectra are grouped by wavelength and averaged

5.3 Stacking procedure

For each input file, the stacker:

1. checks whether the file extension is .fits or .FITS;
2. opens the primary HDU and reads the flux array;
3. reconstructs the wavelength array from CRVAL1, CDELT1, and the number of flux entries;
4. applies the internal redshift correction formula wavelength / (1 + z);
5. removes points with NaN flux or flux equal to zero;
6. creates a specutils Spectrum object;
7. builds a common dispersion grid using the largest wavelength step among all spectra;
8. resamples each spectrum with FluxConservingResampler;
9. combines all resampled spectra and calculates the mean flux for each wavelength.

5.4 Stacker output files

The stacker creates a timestamped output folder in the current working directory. The folder name follows this pattern:

```
stacked_YYYYMMDD-HHMMSS/
```

The stacker writes the following output files:

- **stacked.fits** - the stacked spectrum as a primary-HDU FITS file
- **stacked.png** - a static PNG plot of the stacked spectrum
- **stacked.html** - an interactive HTML plot of the stacked spectrum
- **interactive_stack.html** - an interactive overlay of the individual resampled spectra and the stacked spectrum

- **<input-name>_<number>.png/html** - optional individual plots, written only when "Save individual spectra" is enabled

In v1.0.1, the previously hard-coded CUNIT1="log" header entry for stacked.fits has been removed. The stacker still writes OBJECT="stacked", CRPIX1, CRVAL1, CDELT1, CRDER1=0, and CTYPE1="WAVELENGTH".

> ***Note:*** *The stacker writes a stacked spectrum but does not currently write a stacked error spectrum or S/N spectrum.*

6. Known limitations and cautions

- **Column consistency:** The extractor applies the selected column names to all loaded files. Mixed column naming across files can cause individual files to be skipped.
- **Continuum array shape:** Provided continuum values should have the same length as the flux array after any flattening/interpolation step.
- **STATUS handling:** Rejected points are set to zero rather than removed from the arrays.
- **Stacker units:** The stacker uses fixed units in the GUI call. If your data use different units, adjust the source code before stacking.
- **Stacker errors:** The stacker uses flux arrays only and does not propagate flux-error files.

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